Code Coverage Report

Generated 2026-11-12T16:45:25.898236+00:00. Application source: 6b5464298c949806ebba0b7bd1d903c962b5255d.

Environment: Linux-7.2.14-200.fc44.x86_64-x86_64-with-glibc2.43; g++ (GCC) 16.2.1 20261029 (Red Hat 16.2.1-3); gcovr 8.8.

94.7%: **1476 / 1558** measured lines covered.

Scope and method

GCC/gcov instrumentation with gcovr, restricted to project src/ files and headers; src/main.cpp, tests and downloaded dependencies are excluded. The executable startup, CLI and real TLS path are additionally exercised by the system validation runner but are not included in this coverage percentage. No exclusions for exception branches or unreachable branches are applied. Compiler-generated C++ exception/cleanup edges remain in the branch denominator. No minimum percentage was specified in the five course PDFs.

Negative-hit parse handling matches the existing CI workaround: --gcov-ignore-parse-errors negative_hits.warn_once_per_file. The collection log is included as coverage-collection.log; inspect warnings when regenerating.

Interactive source coverage · Raw coverage JSON · Summary JSON

File	Covered	Total	Percent
src/api/AdminRoutes.cpp	111	115	96.5
src/api/Authorization.h	20	20	100.0
src/api/Routes.cpp	318	330	96.4
src/auth/PasswordHasher.cpp	68	75	90.7
src/auth/PasswordHasher.h	1	1	100.0
src/db/Database.cpp	42	46	91.3
src/db/Schema.cpp	42	45	93.3
src/db/Seed.cpp	24	24	100.0
src/services/AdminTrainService.cpp	187	190	98.4
src/services/AdminTrainService.h	1	1	100.0
src/services/AuditService.cpp	29	29	100.0
src/services/AuditService.h	1	1	100.0
src/services/AuthService.cpp	52	52	100.0
src/services/BookingService.cpp	250	282	88.7
src/services/BookingService.h	2	2	100.0
src/services/TrainService.cpp	148	151	98.0
src/services/TrainService.h	2	2	100.0
src/services/UserService.cpp	67	73	91.8
src/services/UserService.h	1	1	100.0
src/util/Log.cpp	20	21	95.2
src/util/TlsConfig.cpp	13	18	72.2
src/util/TlsConfig.h	1	1	100.0
src/util/Validation.cpp	76	78	97.4

Follow up

Prioritize untested error and rollback paths visible in the HTML report. A passing test suite is not equivalent to complete branch coverage.